

Rotman

RITCxQuestrom Training Session

Liability Trading & Algorithmic Market Making Capstone



Rotman School of Management
UNIVERSITY OF TORONTO

Section	Time
Resource Page	10 min
RIT	45 min
Platform	25 min
API Connection	20 min
Cases	30 min
Liability Trading	15 min
Algo Market Making	15 min
Q&A	5 min

RESOURCE PAGE

Practice session and server
Windows / Mac setup
How to connect
How scoring works

Windows

Desktop
application

[Download the RIT User Application](#)

Run the installer, then open RIT.

Server: flserver.rotman.utoronto.ca

Port: 16650



Rotman Interactive Trader Login

Trader ID:

Password:

Server:

Mac

Browser
or Mac app

[Browser: client.rotmanrit.com:14980](#)

[Or download the Apple Silicon / Intel app.](#)

[Unzip → drag to Applications → open.](#)

Server: flserver.rotman.utoronto.ca

Port: 16655

RIT Market Simulator Web Application
BETA This is a beta test version. Feedback and questions are welcome at financelab@rotmanutoronto.ca

Connect to RIT DMA REST API

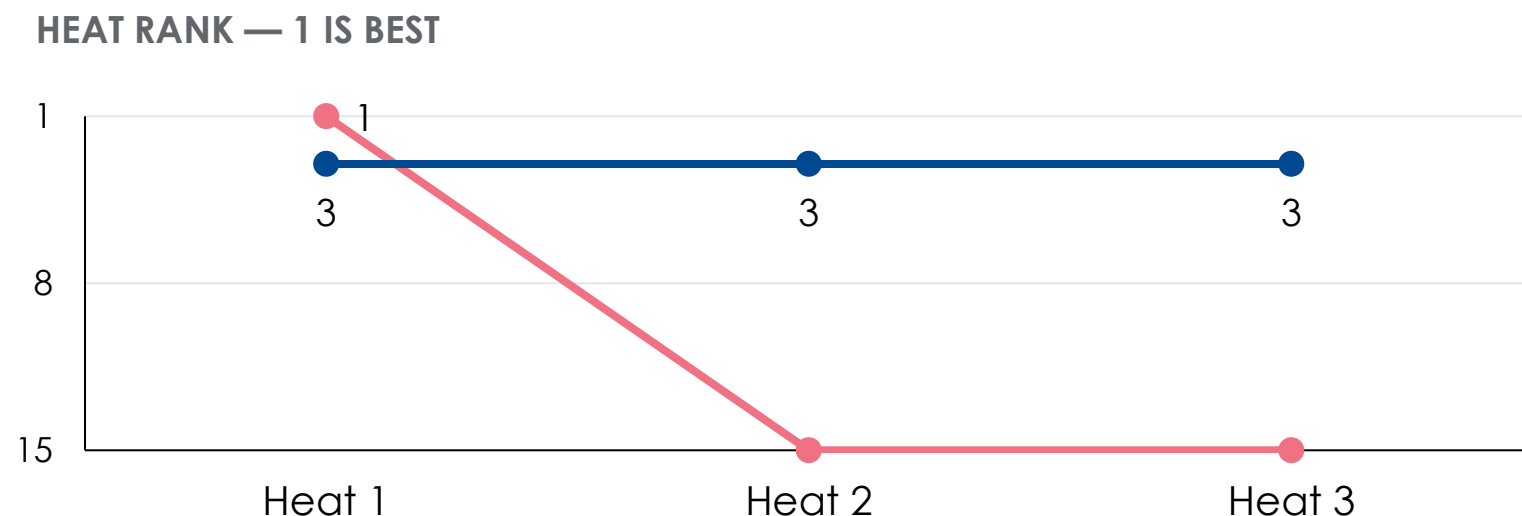
Username

Password

Server
e.g. flserver.rotman.utoronto.ca

Port
e.g. 10001

15 teams. Which team performs better across these three heats?



AVERAGE HEAT RANK

Team A: 10.33

Team B: 3.00

Team B ranks higher.
Lower average is better.

Rank each heat → Average the ranks → Rank each case

Final result: 50% Liability Trading + 50% Market Making

Read the market, place an instruction, check the result

PORTFOLIO

What do I own?

Position, cost, last, bid and ask.

MARKET DEPTH

What can trade now?

Price, quantity and priority.

ORDER ENTRY / BLOTTER

What did I instruct?

Market / limit; live, partial, filled, cancelled.

TRADER INFO / LOG

What changed?

Limits, P&L, transaction costs; save workspace.

DMA REST API

Python → DMA-enabled
simulation server

No desktop client bridge

DMA endpoint + trader login

CLIENT REST API

Python → local RIT client
→ simulation server

Client open and logged in

Local port + matching API key

The screenshot shows a window titled "API Info" with a close button (X) in the top right corner. The window is divided into three sections:

- VBA API**: Documentation can be found [here](#) and requires the companion [Excel RTD Links](#) installation.
- DMA REST API**: Interactive documentation can be found [here](#).
 - Base URL:**
 - Username:**
 - Password:**
 - Authorization Header:**
This field is only used if you are setting the Authorization header manually. Otherwise, supplying the username and password values to your programming language's request library is sufficient.
- Client REST API**: Interactive documentation can be found [here](#).
 - Port:**
 - API Key:**

At the bottom of the window are two buttons: "Save" and "Cancel".

Why not sell the whole block directly into the market?

A large sale can move prices through the depth it uses and the signal it sends.

PRICE IMPACT: DEPTH

The sale uses up the bids

Remaining shares reach lower prices.

PRICE IMPACT: SIGNALS

“Is there more selling to come?”

Other traders may lower bids or reduce size.

Client sells the block to you → You decide how to sell it into the market

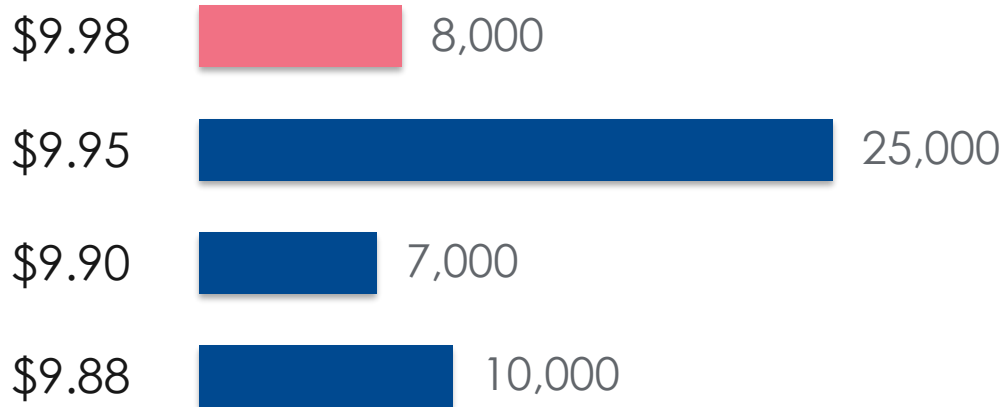
Liability Trading Case

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The first share and the whole block have different prices

50,000 CRZY at \$9.90: the best bid is only the start of the exit.

BIDS USED TO SELL 50,000 SHARES



Bar length = shares available at that price

FIRST SHARE → VWAP

\$9.98 → \$9.9338

Apparent edge → depth-adjusted edge

Liability Trading Case

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Trading faster and waiting longer carry different risks

SELL QUICKLY



More depth consumed now.
Less time exposed to price moves.

SELL OVER TIME



Liquidity may replenish.
Inventory remains at market risk while waiting.

Trading rules and fines: finish the tender trade

Your role is to unwind accepted tenders.

FRONT-RUNNING

**Trading ahead of
a tender**

Trying to profit from its
expected market impact.

Lead to penalties.

UNRELATED SPECULATION

**Taking an extra
price bet**

Buying extra shares to bet on
a price rise.

Lead to penalties.

AT THE CLOSE

**Leaving a long
or short position**

Residual inventory keeps you
exposed to price moves.

A nonzero position is penalized.

Closing check: zero positions + no live orders

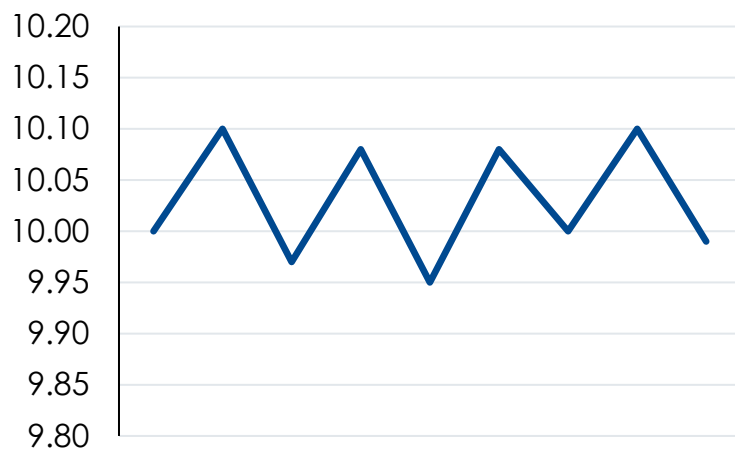
Algorithmic Market Making Capstone

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A spread is an opportunity; inventory connects the fills

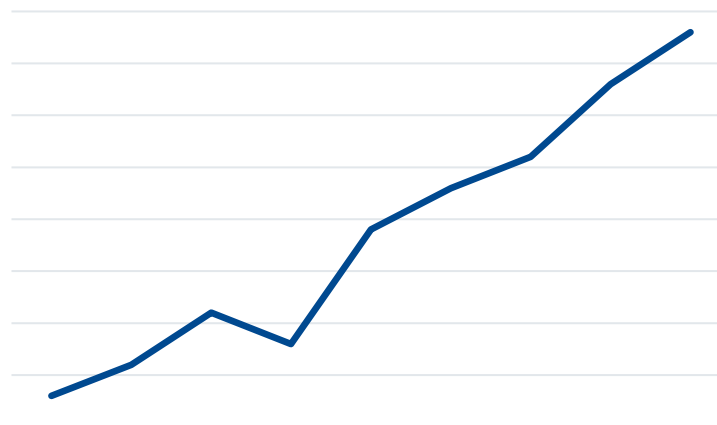
Buy 500 at \$10.00, then sell at \$10.10 → \$50 before costs. What happens in between?

RANGE



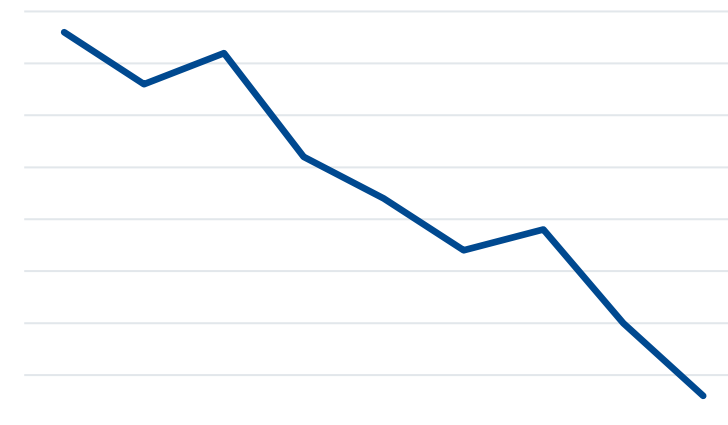
Can both sides fill?

RISING



More sells → short inventory?

FALLING



More buys → long inventory?

How should our quotes respond to inventory?

CNR: our buy fills. We now own 500 shares.



Already long: how would you adjust your next quotes?

Feebates: why pay someone to trade?

Exchanges compete for both quotes and incoming trades.

CNR & AC

Reward providing liquidity

Market maker → receive a rebate

Market taker → pay a fee

Encourage liquidity provision.

RY

Reward taking liquidity

Market maker → pay a fee

Market taker → receive a rebate

Attract incoming trades.

What matters to us: spread, fees / rebates, and inventory risk.

Questions before practice

What would you like to see again?

LIABILITY TRADING

Is the client price enough?

Market risk & Liquidity risk

MARKET MAKING

Can we complete the trade profitably?

One-sided fills build inventory.
Prices, fees and rebates all matter.

Live practice: October 6 | 4–5 p.m. Eastern time